

Supplementary material

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Manuscript source commit `9aa4f05cc96c` — the state of the manuscript and analysis sources from which this supplement was generated. The final package commit and tag are recorded in S17; they differ from the source commit by the metadata-only step that records this provenance, and no byte identity between the two is claimed.

Protocol tag `xgboost-hpo-protocol-v3`. Results tag `xgboost-hpo-confirmatory-results-v1`.

S1. Protocol lineage and tags

tag	commit	what it records
<code>v0.1.0-dissertation</code>	<code>67d9fe5</code>	the historical implementation this study reconstructs
<code>xgboost-hpo-protocol-v1</code>	<code>355c290</code>	the specification frozen before the pilot
<code>xgboost-hpo-protocol-v2</code>	<code>c50c380</code>	the confirmatory protocol after pilot Stage A
<code>xgboost-hpo-protocol-v3</code>	<code>9b15ba7</code>	the final frozen protocol, before any comparative result
<code>xgboost-hpo-confirmatory-results-v1</code>	<code>a4f20ed</code>	the first adversarially verified analysis

The protocol tag records what was specified **before** any comparative result existed; the results tag records the first verified analysis. Neither was moved.

S2. Amendment chronology

The protocol carries **23 numbered amendments**, each recording what caused it and whether any arm result had been observed when it was made. None had. The ledger also records two process failures found by review rather than by the author, four engineering defects caught by a pre-freeze smoke run, and the corrections made to two claims after independent verification refuted the author's version. It is reproduced in full in `PROTOCOL_AMENDMENTS.md`.

S3. Hyperparameter space

hyperparameter	lower	upper	type
<code>subsample</code>	0.05	1.0	continuous
<code>colsample_bytree</code>	0.05	1.0	continuous
<code>colsample_bylevel</code>	0.05	1.0	continuous
<code>learning_rate</code>	0.01	0.3	continuous
<code>max_depth</code>	3	18	integer
<code>gamma</code>	0.05	5.0	continuous
<code>n_estimators</code>	50	700	integer

All seven enter the design in coded units on `[-1, 1]`; integers are cast by `int(round(.))` at evaluation time.

S4. Responses and objective construction

response	role	transform
Accuracy_Mean	quality	none
Precision_Mean	quality	none
Recall_Mean	quality	none
Specificity_Mean	quality	none
RocAuc_Mean	quality	none
LogLoss_Mean	quality	none
Leaves_Mean	cost	log1p

S4.1 Frozen factor models

dataset	fitting sample	audit rows used	quality weights	Kaiser	λ_3/λ_4
MAGIC	88 design rows	0	0.5556, 0.4444	2	2.93
Spambase	88 design rows	0	0.7749, 0.2251	2	2.10
Adult	88 design rows	0	0.5470, 0.4530	1	1.34
Bank Marketing	88 design rows	0	0.5875, 0.4125	1	1.06

One model per dataset, fitted on the 88 design rows only and applied to all 30 replications. The 78-point audit-only external construction is excluded from the fit (amendment 23): 64 of its 78 coded points are identical to the external validation set and the rest are the same axial runs.

S5. Panel screening and dataset roles

dataset	C1 raw conflict	C2 front / curvature	C3 min R ²	C4 range	role
MAGIC	-0.2509	7 / 0.1729	0.6913	2706.3×	primary_geometry_confirmatory
Spambase	-0.4621	2 / undefined	0.8987	591.2×	boundary_geometry_control
Adult	-0.3977	6 / 0.1992	0.9078	3421.3×	primary_geometry_confirmatory
Bank Marketing	-0.4100	6 / 0.1983	0.8700	4355.5×	primary_geometry_confirmatory

Roles were assigned from these pre-campaign measurements and never revised from an optimizer outcome.

S6. Method and control registry

identifier	kind	runner stage	standalone budget	revalidated	isolates
HISTORICAL-WS-asrun	arm	historical_ws_asrun	108	yes	the dissertation's pipeline reproduced bit-faithfully by calling the frozen v0.1.0-dissertation solver: maximization orientation, uncoded natural-unit surfaces, observed-extrema normalization, asymmetric 20-point grid with no pure-quality vertex, and no gate. Reproduced, not repaired.
HISTORICAL-WS	arm	historical_ws	186	yes	the same weighted sum, solver, symmetric grid and surrogates as WS-S, differing ONLY in the normalization reference. WS-S to HISTORICAL-WS is therefore a single-factor contrast on normalization.
WS-S	arm	ws_s	186	yes	weighted-sum scalarization over the surrogate payoff reference. The geometry baseline for the primary contrast.
NBI-S	arm	nbi_s	186	yes	Normal Boundary Intersection on the same surrogates and the same surrogate reference as WS-S. WS-S to NBI-S isolates front-construction geometry.
NBI-R	arm	nbi_r	386	yes	NBI with anchors and payoff matrix from direct search on the REAL objectives. NBI-S to NBI-R isolates anchor and payoff provenance.
ANCHOR-INJECTION-CONTROL	control	anchor_injection_control	—	yes	NBI-S's own revalidated candidate set augmented with the same empirical anchors NBI-R receives, changing nothing else. Separates the part of any NBI-S to NBI-R gap that is set composition from the part that is relocated geometry. Costs no new real evaluations.
GRID	baseline	direct_baselines	—	n/a	direct-search comparator at the frozen 386-evaluation comparator budget, which is matched to the most expensive arm (NBI-R) and not pairwise to each arm
RANDOM	baseline	direct_baselines	—	n/a	direct-search comparator at the frozen 386-evaluation comparator budget, matched to

identifier	kind	runner stage	standalone budget	revalidated	isolates
					the most expensive arm and not pairwise to each arm
BAYES-QUALITY	baseline	direct_baselines	—	n/a	single-objective Bayesian optimization on quality, at the frozen comparator budget; excluded from front indicators and from the augmented reference
BAYES-COST	baseline	direct_baselines	—	n/a	single-objective Bayesian optimization on cost; same exclusion
TPE-QUALITY	baseline	direct_baselines	—	n/a	single-objective TPE on quality; same exclusion
TPE-COST	baseline	direct_baselines	—	n/a	single-objective TPE on cost; same exclusion
NSGA2-MATCHED	baseline	direct_baselines	—	n/a	NSGA-II at $32 \times 12 = 384$ evaluations, run at the frozen comparator budget to within a disclosed two-evaluation shortfall (384 against 386); this is a budget-level match to the most expensive arm, not a pairwise match to NBI-S
NSGA2-UNMATCHED	context	nsga2_unmatched	—	n/a	NSGA-II at 10x the matched budget, one replication per dataset. A CONTEXT baseline: no fairness claim attaches to it and it enters no budget-matched comparison.
DESIGN	shared stage	design	—	n/a	88 face-centred central composite runs
EXTERNAL-VALIDATION-AUDIT	shared stage, AUDIT-ONLY	external_validation	—	n/a	78 points: the design's complementary half fraction plus 14 axial runs at half radius. Reaches the reliability gate diagnostics and NOTHING else. A gate failure changes no execution.
SURROGATE-ANCHORS	reference construction	surrogate_anchors	—	n/a	per-objective minimization of the fitted surrogates over the coded box
EMPIRICAL-ANCHORS	reference construction	empirical_anchors	—	n/a	direct search on the REAL objectives, one budget per objective. Best found within the declared budget; never described as certified optima.
REFERENCE-CORE	reference construction	reference_core	—	n/a	the reference front used for indicators
AUGMENTED-REFERENCE	reference construction	augmented_reference	—	n/a	reports self_grading_share_of_front

identifier	kind	runner stage	standalone budget	revalidated	isolates
HOLDOUT- CONFIRMATION	confirmation stage	holdout_confirmation	—	n/a	selected candidates re-measured on the held-out partition

S7. Budget

method	stage	logical per unit
anchor_injection_control	candidate_validation	2
bayes_cost	direct_search	386
bayes_quality	direct_search	386
design	design	88
empirical_anchor_search	anchor	200
external_validation_audit	external_audit	78
grid	direct_search	386
historical_ws_asrun_revalidation	candidate_validation	20
historical_ws_revalidation	candidate_validation	20
holdout_confirmation	holdout_audit	5
nbi_r_revalidation	candidate_validation	20
nbi_s_revalidation	candidate_validation	20
nsga2	direct_search	384
random	direct_search	386
tpe_cost	direct_search	386
tpe_quality	direct_search	386
ws_s_revalidation	candidate_validation	20
total		3,173

Of which 83 audit-only and 3,090 solution-producing.

quantity	value
logical evaluations per unit	3,173
confirmatory units	120
unmatched NSGA-II, outside the unit	15,360
campaign total	396,120
of which audit-only	9,960
unique physical fits	377,316

The campaign total already contains the unmatched NSGA-II charge, inside the four replication-0 ledgers. Adding it again gives 411,480 and is wrong.

S7.1 Standalone cost per arm

arm	standalone logical
HISTORICAL-WS-asrun	108
HISTORICAL-WS	186
WS-S	186
NBI-S	186
NBI-R	386
each direct-search comparator	386

The comparator budget is the **maximum over arms** (NBI-R). A direct-search comparator therefore receives about twice the real evaluations WS-S and NBI-S require standalone, and any comparison against them must say so.

S8. Randomness

Seeds derive from BLAKE2b over the namespace `xgboost-hpo-vrfnbi/v3` and the tuple (dataset, replication, method, stage), fed to `numpy.random.SeedSequence`.

dataset	unit seed, replication 0	example method seed
MAGIC	20261914	12501833259513610416
Spambase	20262914	1639200741726807977
Adult	20263914	3386437749439664214
Bank Marketing	20264914	7633668092872975921

All 3,840 campaign streams are distinct and stable across processes and `PYTHONHASHSEED`.

S9. Primary statistics, both references

S9.1 CORE reference (primary)

dataset	contrast	median Δ	95% CI	W/T/L	rank- biserial	Holm p	Nadeau–Bengio p
MAGIC	HISTORICAL-WS -> WS-S	+0.0000	[-0.0022, +0.0056]	14/3/13	+0.14	0.5165	0.6697
MAGIC	WS-S -> NBI-S	+0.0374	[+0.0144, +0.0687]	26/0/4	+0.92	9.425e-07	0.0860
MAGIC	NBI-S -> NBI-R	-0.0741	[-0.0957, -0.0482]	2/0/28	-0.99	2.794e-08	0.0254
Adult	HISTORICAL-WS -> WS-S	-0.0006	[-0.0033, +0.0000]	9/4/17	-0.28	0.2087	0.7467
Adult	WS-S -> NBI-S	+0.2549	[+0.1740, +0.5614]	30/0/0	+1.00	5.588e-09	0.0148
Adult	NBI-S -> NBI-R	-0.0192	[-0.1610, +0.0299]	11/0/19	-0.37	0.1607	0.4161
Bank Marketing	HISTORICAL-WS -> WS-S	+0.0000	[-0.0088, +0.0000]	8/8/14	-0.21	0.3896	0.8893
Bank Marketing	WS-S -> NBI-S	+0.2281	[+0.1670, +0.4741]	24/0/6	+0.88	9.717e-06	0.0764
Bank Marketing	NBI-S -> NBI-R	-0.4093	[-0.5821, -0.2059]	7/0/23	-0.82	4.733e-05	0.0791
Spambase	HISTORICAL-WS -> WS-S	+0.0018	[+0.0000, +0.0398]	16/7/7	+0.41	0.1175	0.6846
Spambase	WS-S -> NBI-S	+0.0500	[-0.0043, +0.4369]	18/2/10	+0.41	0.1175	0.6550
Spambase	NBI-S -> NBI-R	-0.6932	[-0.9839, -0.5217]	5/0/25	-0.78	0.0005664	0.2763

The Nadeau–Bengio column is the pre-declared corrected resampled t , at an inflation of $\sqrt{8.5} = 2.9155$. It is a sensitivity, not the primary test, and is reported in full including where it is adverse to the finding.

S9.2 AUGMENTED reference (mandatory sensitivity)

dataset	contrast	median Δ	95% CI	W/T/L	rank-biserial	Holm p	Nadeau–Bengio p
MAGIC	HISTORICAL-WS -> WS-S	+0.0000	[-0.0029, +0.0047]	13/3/14	+0.05	0.8288	0.7807
MAGIC	WS-S -> NBI-S	+0.0382	[+0.0179, +0.0668]	27/0/3	+0.95	2.049e-07	0.0927
MAGIC	NBI-S -> NBI-R	-0.0794	[-0.0939, -0.0451]	1/0/29	-0.99	2.794e-08	0.0286
Adult	HISTORICAL-WS -> WS-S	-0.0006	[-0.0018, +0.0000]	8/4/18	-0.23	0.3158	0.7431
Adult	WS-S -> NBI-S	+0.1936	[+0.1322, +0.3717]	30/0/0	+1.00	5.588e-09	0.0204
Adult	NBI-S -> NBI-R	-0.0535	[-0.1181, +0.0035]	10/0/20	-0.54	0.01741	0.3756
Bank Marketing	HISTORICAL-WS -> WS-S	-0.0001	[-0.0021, +0.0000]	7/8/15	-0.23	0.3382	0.7957
Bank Marketing	WS-S -> NBI-S	+0.1792	[+0.1074, +0.2045]	25/0/5	+0.94	7.655e-07	0.0643
Bank Marketing	NBI-S -> NBI-R	-0.1699	[-0.2369, -0.0772]	7/0/23	-0.82	4.733e-05	0.1109
Spambase	HISTORICAL-WS -> WS-S	+0.0000	[+0.0000, +0.0096]	14/11/5	+0.48	0.1283	0.6545
Spambase	WS-S -> NBI-S	+0.0503	[+0.0000, +0.2079]	19/2/9	+0.33	0.1283	0.6531
Spambase	NBI-S -> NBI-R	-0.3464	[-0.4880, -0.2553]	4/1/25	-0.84	0.0002382	0.0658

The Nadeau–Bengio column is the pre-declared corrected resampled t, at an inflation of $\sqrt{8.5} = 2.9155$. It is a sensitivity, not the primary test, and is reported in full including where it is adverse to the finding.

S10. Secondary indicators

dataset	entity	HV ratio	IGD ⁺	GD	spacing	joint-ND	front size
MAGIC	HISTORICAL-WS-asrun	0.7627	0.1477	0.1174	0.0460	0.1458	8.0000
MAGIC	HISTORICAL-WS	0.8613	0.1028	0.1153	0.0890	0.5798	10.0000
MAGIC	WS-S	0.8614	0.0930	0.1013	0.0862	0.5857	9.0000
MAGIC	NBI-S	0.9430	0.0511	0.0781	0.0496	0.5917	11.0000
MAGIC	NBI-R	0.8588	0.0856	0.1015	0.0244	0.6667	9.0000
MAGIC	ANCHOR-INJECTION-CONTROL	0.9607	0.0444	0.0713	0.0979	0.6259	12.5000
MAGIC	GRID	1.0068	0.0036	0.0393	0.0706	0.9181	17.0000
MAGIC	RANDOM	0.8907	0.0629	0.0775	0.0382	0.4416	13.0000
MAGIC	NSGA2-MATCHED	0.9580	0.0373	0.0858	0.0188	0.6883	30.5000
Spambase	HISTORICAL-WS-asrun	0.0221	0.6561	0.7892	0.1642	0.0000	4.0000
Spambase	HISTORICAL-WS	0.8583	0.1312	0.1837	0.1445	0.8750	3.0000
Spambase	WS-S	0.8819	0.1054	0.2042	0.2049	1.0000	3.0000
Spambase	NBI-S	1.0812	0.0877	0.2343	0.2142	0.9444	4.0000
Spambase	NBI-R	0.0793	0.6200	0.5881	0.0000	1.0000	3.0000
Spambase	ANCHOR-INJECTION-CONTROL	1.0812	0.0877	0.2343	0.2142	0.9444	4.0000
Spambase	GRID	1.0192	0.0000	0.0394	0.2372	1.0000	5.0000
Spambase	RANDOM	0.0000	1.3453	1.3795	0.0256	0.0000	2.0000
Spambase	NSGA2-MATCHED	0.4164	0.4738	0.5878	0.1166	0.2500	5.0000
Adult	HISTORICAL-WS-asrun	0.5196	0.3404	0.2414	0.0176	0.0000	10.5000
Adult	HISTORICAL-WS	0.5422	0.1847	0.1483	0.0930	0.6970	7.5000
Adult	WS-S	0.6028	0.1688	0.1785	0.0890	0.7321	7.0000
Adult	NBI-S	0.9951	0.0579	0.1077	0.1329	0.8258	8.5000
Adult	NBI-R	0.9098	0.1201	0.1374	0.0503	0.7071	9.0000
Adult	ANCHOR-INJECTION-CONTROL	0.9951	0.0578	0.1397	0.1483	0.7386	10.0000
Adult	GRID	1.0476	0.0000	0.0465	0.0947	1.0000	8.5000
Adult	RANDOM	0.4369	0.3334	0.2441	0.0984	0.0000	5.0000
Adult	NSGA2-MATCHED	0.8650	0.1228	0.1821	0.0359	0.4715	28.0000
Bank Marketing	HISTORICAL-WS-asrun	0.2743	0.4275	0.2907	0.0784	0.0000	6.0000
Bank Marketing	HISTORICAL-WS	0.6082	0.1946	0.2087	0.1804	0.6333	7.0000
Bank Marketing	WS-S	0.6010	0.1948	0.2320	0.1862	0.5000	7.0000
Bank Marketing	NBI-S	0.9979	0.0915	0.2173	0.1219	0.6111	9.0000
Bank Marketing	NBI-R	0.5330	0.3582	0.2468	0.0505	0.2917	6.0000
Bank Marketing	ANCHOR-INJECTION-CONTROL	0.9979	0.0898	0.1754	0.1358	0.6667	8.5000
Bank Marketing	GRID	1.0625	0.0000	0.0481	0.1432	1.0000	8.0000
Bank Marketing	RANDOM	0.3190	0.3257	0.3119	0.1037	0.0000	7.0000
Bank Marketing	NSGA2-MATCHED	0.9674	0.0934	0.2182	0.0451	0.4929	20.5000

Undefined spacing. 140 cells are non-finite — spacing and spacing_cv on the 70 method-by-reference blocks whose front has one point. Schott spacing requires at least two gaps, so NaN is the correct value and those units are excluded from spacing summaries rather than propagated.

S11. Surrogate-gate regimes

dataset	quality gate	cost gate	both	role
MAGIC	8/30	100%	27%	primary_geometry_confirmatory
Spambase	9/30	100%	30%	boundary_geometry_control
Adult	0/30	73%	0%	primary_geometry_confirmatory
Bank Marketing	0/30	90%	0%	primary_geometry_confirmatory

The gate is diagnostic, never adaptive: every arm ran at every replication whatever the gate said.

S12. Controls

S12.1 Anchor-injection control

dataset	injection effect	full NBI-S→NBI-R gap	CHIM extent ratio
MAGIC	+0.0133	-0.0741	0.811
Spambase	+0.0000	-0.6932	0.370
Adult	+0.0000	-0.0192	0.204
Bank Marketing	+0.0000	-0.4093	0.272

The control set is a superset of NBI-S's, so its effect is non-negative by construction. Within-dataset Spearman between the per-replication CHIM extent ratio and the hypervolume gap: MAGIC -0.11, Spambase +0.32, Adult +0.68, Bank Marketing +0.68. Association, not cause, and absent on MAGIC.

S12.2 Solver health, NBI-S against NBI-R

dataset	certified (S/R)	max equality residual (S/R)	distinct configs (S/R)
MAGIC	1.000 / 1.000	8.6e-10 / 9.3e-10	20 / 20
Spambase	1.000 / 1.000	5.9e-10 / 5.8e-10	20 / 20
Adult	1.000 / 1.000	4.4e-10 / 5.6e-10	20 / 20
Bank Marketing	1.000 / 1.000	6.1e-10 / 6.8e-10	20 / 20

Identical on every dataset, so the deficit is not solver failure, rounding or candidate collapse.

S12.3 Historical reconstruction

dataset	as-run	shared specification	WS-S	$\Delta(\text{shared} - \text{as-run})$
MAGIC	0.7627	0.8613	0.8614	+0.0994
Spambase	0.0221	0.8583	0.8819	+0.5404
Adult	0.5196	0.5422	0.6028	+0.1480
Bank Marketing	0.2743	0.6082	0.6010	+0.3126

S13. Baselines

dataset	GRID	RANDOM	NSGA-II matched	WS-S	NBI-S	NBI-S wins vs GRID
MAGIC	1.0068	0.8907	0.9580	0.8614	0.9430	0/30
Spambase	1.0192	0.0000	0.4164	0.8819	1.0812	12/30
Adult	1.0476	0.4369	0.8650	0.6028	0.9951	5/30
Bank Marketing	1.0625	0.3190	0.9674	0.6010	0.9979	10/30

The unmatched NSGA-II run receives ten times the matched budget on one replication per dataset and is a **context baseline**: no fairness claim attaches to it and it enters no budget-matched comparison and neither reference.

S14. Holdout confirmation

dataset	arm	internal	holdout	median paired drop
MAGIC	historical_ws	0.8580	0.8604	-0.0017
MAGIC	historical_ws_asrun	0.8477	0.8541	-0.0034
MAGIC	nbi_r	0.8587	0.8603	-0.0013
MAGIC	nbi_s	0.8593	0.8594	-0.0015
MAGIC	ws_s	0.8598	0.8628	-0.0016
Spambase	historical_ws	0.8871	0.8947	-0.0059
Spambase	historical_ws_asrun	0.9046	0.9088	-0.0044
Spambase	nbi_r	0.9185	0.9235	-0.0117
Spambase	nbi_s	0.8898	0.9072	-0.0110
Spambase	ws_s	0.8890	0.8990	-0.0076
Adult	historical_ws	0.8408	0.8392	+0.0016
Adult	historical_ws_asrun	0.8412	0.8406	+0.0004
Adult	nbi_r	0.8433	0.8430	-0.0018
Adult	nbi_s	0.8437	0.8445	-0.0012
Adult	ws_s	0.8409	0.8415	+0.0020
Bank Marketing	historical_ws	0.8985	0.8970	+0.0009
Bank Marketing	historical_ws_asrun	0.8975	0.8974	+0.0011
Bank Marketing	nbi_r	0.8984	0.8990	-0.0002
Bank Marketing	nbi_s	0.8986	0.8973	+0.0012
Bank Marketing	ws_s	0.8988	0.8973	+0.0014

All magnitudes are below 0.012 and several are negative, meaning the held-out partition scored better than the internal resampling. This is read as the absence of gross selection optimism, **not** as a ranking of arms.

S15. Finite-reference diagnostics

dataset	rows with HV ratio > 1	share	maximum
MAGIC	36/270	13%	1.057
Spambase	95/270	35%	9.912
Adult	76/270	28%	1.176
Bank Marketing	93/270	34%	2.349

Overall 27.8% of rows exceed 1, maximum 9.912. The core reference is a finite method-independent set of 288 points, not the true Pareto front; a ratio above 1 means the candidate set improved on that finite reference.

S16. Claims and evidence

The manuscript's headline numerical claims, each with its estimand, source artifact, verification status, allowed wording and prohibited stronger wording, are in `CONFIRMATORY_CLAIMS_AND_EVIDENCE.md`. Two are recorded in corrected form because independent verification refuted the author's version.

S17. Reproducibility manifest

item	value
manuscript source commit	9aa4f05cc96c06105293cfa46af3729635025c3c
final package tag	paper2-manuscript-v3
relationship	the package commit adds only this provenance metadata and the compiled PDFs; no manuscript text, analysis artifact or number differs
protocol tag	xgboost-hpo-protocol-v3
results tag	xgboost-hpo-confirmatory-results-v1
datasets	MAGIC, Spambase, Adult, Bank Marketing
replications	30 per dataset, 120 units
primary endpoint	hv_ratio against the core reference
campaign runtime	9 h 46 min, 14 workers × 1 thread

Raw datasets and evaluation caches are deliberately unversioned; the design, the frozen factor models, every analysis artifact and every script are committed.